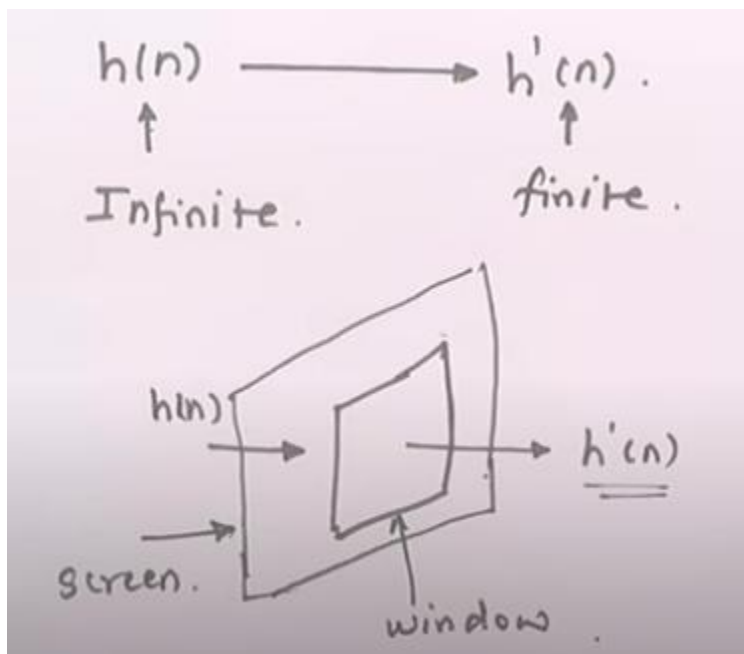


Design Techniques for FIR Filters

The well-known methods of designing FIR filters are as follows;

1. Fourier Series method
2. Window method
3. Frequency sampling method
4. Optimum filter design method

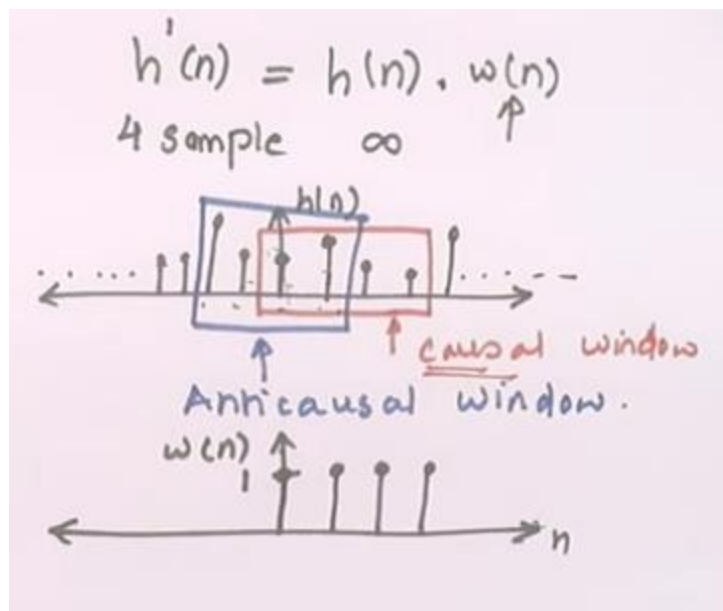
Windowing Technique for FIR Filter Design



$$h'(n) = h(n) \cdot w(n)$$

4 sample ∞ $\uparrow$

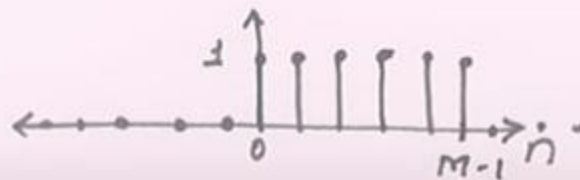
The diagram shows a discrete-time signal $h(n)$ plotted on a horizontal axis. The signal is represented by a series of vertical bars of varying heights. The signal is labeled $h(n)$ and is shown to be infinite in duration, indicated by the ∞ symbol. The signal is also labeled '4 sample'.



Types of Windows

1.] Rectangular window.

$$w_R(n) = \begin{cases} 1 & 0 \leq n \leq M-1 \\ 0 & \text{otherwise} \end{cases}$$



2.] Hamming window.

$$w_H(n) = \begin{cases} 0.54 + 0.46 \cos\left(\frac{2\pi n}{M-1}\right) & 0 \leq n \leq M-1 \\ 0 & \text{otherwise} \end{cases}$$

3.] Hanning.

$$W_{Hn}(n) = \begin{cases} 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right) & 0 \leq n \leq M-1 \\ 0 & \text{otherwise.} \end{cases}$$

4.] Blackman window.

$$W_B(n) = 0.42 + 0.5 \cos\left(\frac{2\pi n}{M-1}\right) + 0.08 \cos\left(\frac{4\pi n}{M-1}\right)$$

5.] Bartlett window.

$$W_{BT}(n) = 1 - \frac{2|n|}{M-1} \quad |n| \leq M-1$$

9.6 DESIGN OF FIR FILTERS USING WINDOWS

The procedure for designing FIR filter using windows is:

1. Choose the desired frequency response of the filter $H_d(\omega)$.
2. Take inverse Fourier transform of $H_d(\omega)$ to obtain the desired impulse response $h_d(n)$.
3. Choose a window sequence $w(n)$ and multiply $h_d(n)$ by $w(n)$ to convert the infinite duration impulse response to a finite duration impulse response $h(n)$.
4. The transfer function $H(z)$ of the filter is obtained by taking Z-transform of $h(n)$.

9.6.1 Rectangular Window

The weighting function (window function) for an N -point rectangular window is given by

$$w_R(n) = \begin{cases} 1, & -\frac{(N-1)}{2} \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{elsewhere} \end{cases} \quad \text{or} \quad w_R(n) = \begin{cases} 1, & 0 \leq n \leq (N-1) \\ 0, & \text{elsewhere} \end{cases}$$

The spectrum (frequency response) of rectangular window $W_R(\omega)$ is given by the Fourier transform of $w_R(n)$.

$$\begin{aligned}
 W_R(\omega) &= \sum_{n=-(N-1)/2}^{(N-1)/2} e^{-j\omega n} = \sum_{n=0}^{N-1} e^{-j\omega \left(n - \frac{N-1}{2}\right)} \\
 &= \sum_{n=0}^{N-1} e^{-j\omega n} e^{j\omega \frac{N-1}{2}} = e^{j\omega \frac{N-1}{2}} \sum_{n=0}^{N-1} e^{-j\omega n} \\
 &= e^{j\omega \left(\frac{N-1}{2}\right)} \left[\frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}} \right] \\
 &= e^{j\frac{\omega N}{2}} e^{-j\frac{\omega}{2}} \frac{e^{-j\frac{\omega N}{2}} \left[e^{j\frac{\omega N}{2}} - e^{-j\frac{\omega N}{2}} \right]}{e^{-j\frac{\omega}{2}} \left[e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}} \right]} \\
 &= \frac{e^{j\frac{\omega N}{2}} - e^{-j\frac{\omega N}{2}}}{e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}} = \frac{\sin \frac{\omega N}{2}}{\sin \frac{\omega}{2}}
 \end{aligned}$$

The frequency spectrum for $N = 31$ is shown in Figure 9.7. The spectrum $W_R(\omega)$ has two features that are important. They are the width of the main lobe and the side lobe amplitude. The frequency response is real and its zero occurs when $\omega = 2k\pi/N$ where k is an integer. The response for ω between $-2\pi/N$ and $2\pi/N$ is called the main lobe and the other lobes are called side lobes. For rectangular window the width of main lobe is $4\pi/N$. The first side lobe will be 13 dB down the peak of the main lobe and the roll off will be at 20 dB/decade. As the window is made longer, the main lobe becomes narrower and higher, and the side lobes become more concentrated around $\omega = 0$, but the amplitude of side lobes is unaffected. So increase in length does not reduce the amplitude of ripples, but increases the frequency when rectangular window is used.

If we design a low-pass filter using rectangular window, we find that the frequency response differs from the desired frequency response in many ways. It does not follow quick transitions in the desired response. The desired response of a low-pass filter changes abruptly from pass band to stop band, but the actual frequency response changes slowly. This region of gradual change is called filter's transition region, which is due to the convolution of the desired response with the window response's main lobe. The width of the transition region depends on the width of the main lobe. As the filter length N increases, the main lobe becomes narrower decreasing the width of the transition region.

The convolution of the desired response and the window response's side lobes gives rise to the ripples in both the pass band and stop band. The amplitude of the ripples is

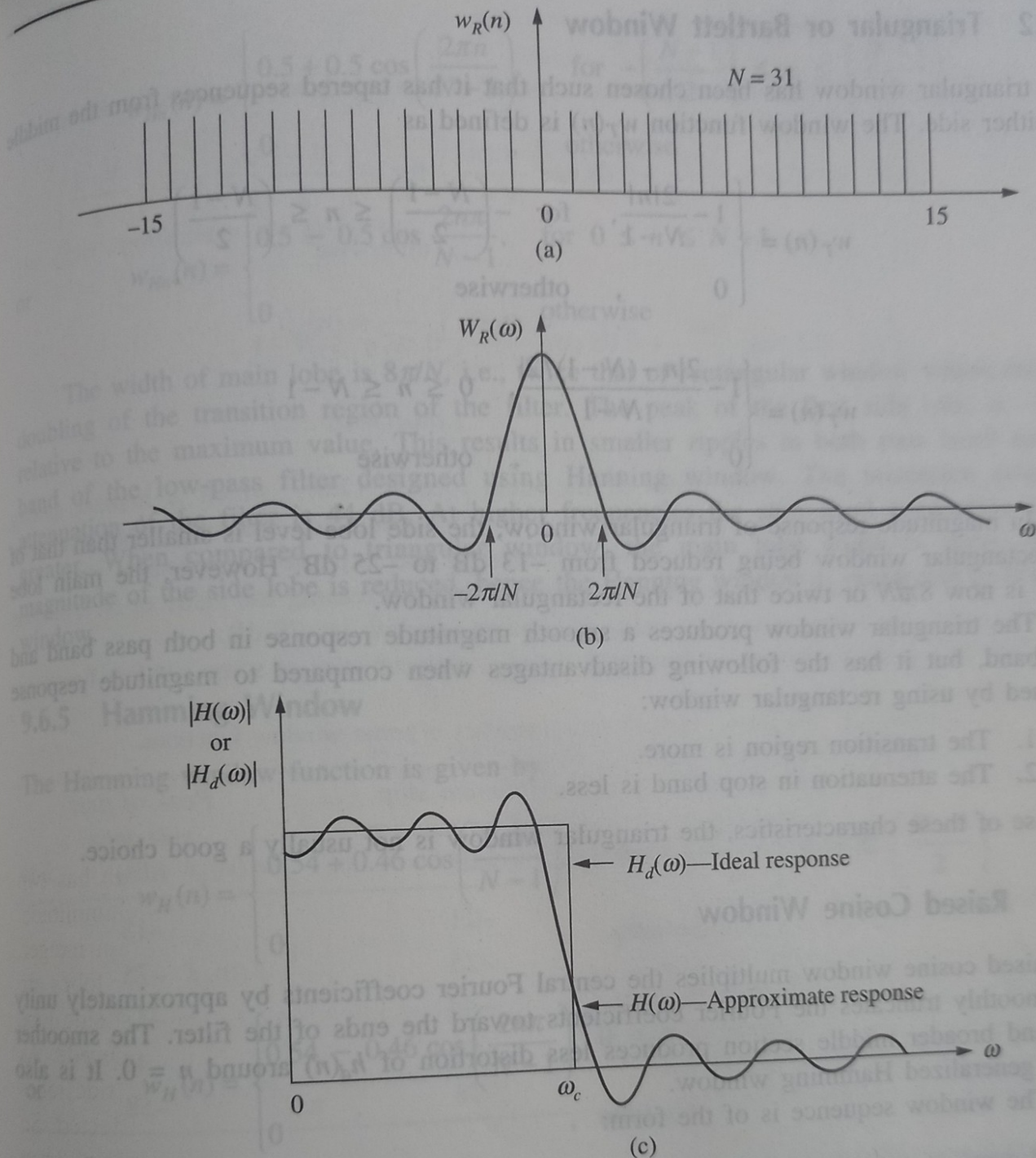


Figure 9.7 (a) Rectangular window sequence, (b) Magnitude response of rectangular window, (c) Magnitude response of low-pass filter approximated using rectangular window.

dictated by the amplitude of the side lobes. This effect, where maximum ripple occurs just before and just after the transition band, is known as Gibbs's phenomenon.

The Gibbs phenomenon can be reduced by using a less abrupt truncation of filter coefficients. This can be achieved by using a window function that tapers smoothly towards zero at both ends.

9.6.2 Triangular or Bartlett Window

The triangular window has been chosen such that it has tapered sequences from the middle on either side. The window function $w_T(n)$ is defined as

$$w_T(n) = \begin{cases} 1 - \frac{2|n|}{N-1}, & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{otherwise} \end{cases}$$

or

$$w_T(n) = \begin{cases} 1 - \frac{2|n - (N-1)/2|}{N-1}, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

In magnitude response of triangular window, the side lobe level is smaller than that of the rectangular window being reduced from -13 dB to -25 dB. However, the main lobe width is now $8\pi/N$ or twice that of the rectangular window.

The triangular window produces a smooth magnitude response in both pass band and stop band, but it has the following disadvantages when compared to magnitude response obtained by using rectangular window:

1. The transition region is more.
2. The attenuation in stop band is less.

Because of these characteristics, the triangular window is not usually a good choice.

9.6.3 Raised Cosine Window

The raised cosine window multiplies the central Fourier coefficients by approximately unity and smoothly truncates the Fourier coefficients toward the ends of the filter. The smoother ends and broader middle section produces less distortion of $h_d(n)$ around $n = 0$. It is also called generalized Hamming window.

The window sequence is of the form:

$$w_H(n) = \begin{cases} \alpha + (1 - \alpha) \cos\left(\frac{2\pi n}{N-1}\right), & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{elsewhere} \end{cases}$$

9.6.4 Hanning Window

The Hanning window function is given by

$$w_{Hn}(n) = \begin{cases} 0.5 + 0.5 \cos\left(\frac{2\pi n}{N-1}\right), & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{otherwise} \end{cases}$$

or

$$w_{Hn}(n) = \begin{cases} 0.5 - 0.5 \cos\frac{2n\pi}{N-1}, & \text{for } 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

The width of main lobe is $8\pi/N$, i.e., twice that of rectangular window which results in doubling of the transition region of the filter. The peak of the first side lobe is -32 dB relative to the maximum value. This results in smaller ripples in both pass band and stop band of the low-pass filter designed using Hanning window. The minimum stop band attenuation of the filter is 44 dB. At higher frequencies the stop band attenuation is even greater. When compared to triangular window, the main lobe width is same, but the magnitude of the side lobe is reduced, hence the Hanning window is preferable to triangular window.

9.6.5 Hamming Window

The Hamming window function is given by

$$w_H(n) = \begin{cases} 0.54 + 0.46 \cos\left(\frac{2\pi n}{N-1}\right), & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{otherwise} \end{cases}$$

or

$$w_H(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2n\pi}{N-1}\right), & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

In the magnitude response for $N = 31$, the magnitude of the first side lobe is down by about 41 dB from the main lobe peak, an improvement of 10 dB relative to the Hanning window. But this improvement is achieved at the expense of the side lobe magnitudes at higher frequencies, which are almost constant with frequency. The width of the main lobe is $8\pi/N$. In the magnitude response of low-pass filter designed using Hamming window, the first side lobe peak is -51 dB, which is -7 dB lesser with respect to the Hanning window filter. However, at higher frequencies, the stop band attenuation is low when compared to that of Hanning window. Because the Hamming window generates lesser oscillations in the side lobes than the Hanning window for the same main lobe width, the Hamming window is generally preferred.

9.6.6 Blackman Window

The Blackman window function is another type of cosine window and given by the equation

$$w_B(n) = \begin{cases} 0.42 + 0.5 \cos \frac{2\pi n}{N-1} + 0.08 \cos \frac{4\pi n}{N-1}, & \text{for } -\left(\frac{N-1}{2}\right) \leq n \leq \left(\frac{N-1}{2}\right) \\ 0, & \text{otherwise} \end{cases}$$

$$\text{or } w_B(n) = \begin{cases} 0.42 - 0.5 \cos \frac{2n\pi}{N-1} + 0.08 \cos \frac{4n\pi}{N-1}, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

In the magnitude response, the width of the main lobe is $12\pi/N$, which is highest among windows. The peak of the first side lobe is at -58 dB and the side lobe magnitude decreases with frequency. This desirable feature is achieved at the expense of increased main lobe width. However, the main lobe width can be reduced by increasing the value of N . The side lobe attenuation of a low-pass filter using Blackman window is -78 dB.

Table 9.2 gives the important frequency domain characteristics of some window functions.

TABLE 9.2 Frequency domain characteristics of some window functions.

Type of window	Approximate transition width of main lobe	Minimum stop band attenuation (dB)	Peak of first side lobe (dB)
Rectangular	$4\pi/N$	-21	-13
Bartlett	$8\pi/N$	-25	-25
Hanning	$8\pi/N$	-44	-31
Hamming	$8\pi/N$	-51	-41
Blackmann	$12\pi/N$	-78	-58

EXAMPLE 9.6 Design an ideal low-pass filter with $N = 11$ with a frequency response

$$H_d(e^{j\omega}) = \begin{cases} 1, & \text{for } -\frac{\pi}{2} \leq \omega \leq \frac{\pi}{2} \\ 0, & \text{for } \frac{\pi}{2} \leq |\omega| \leq \pi \end{cases}$$

Solution: For the given desired frequency response,

$$H_d(\omega) = \begin{cases} 1, & \text{for } -\frac{\pi}{2} \leq \omega \leq \frac{\pi}{2} \\ 0, & \text{for } \frac{\pi}{2} \leq |\omega| \leq \pi \end{cases}$$

The filter coefficients are given by

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} (1) e^{j\omega n} d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{1}{n\pi} \left[\frac{e^{j\frac{n\pi}{2}} - e^{-j\frac{n\pi}{2}}}{2j} \right]$$

$$= \frac{1}{n\pi} \sin \frac{n\pi}{2} \quad \text{for } n \neq 0$$

and $h_d(n) = \frac{1}{2}$ for $n = 0$ [using L'Hospital rule]

$$\therefore h_d(0) = \frac{1}{2},$$

$$h_d(1) = \frac{1}{\pi} \sin \frac{\pi}{2} = \frac{1}{\pi} = h_d(-1)$$

$$h_d(3) = \frac{1}{3\pi} \sin \frac{3\pi}{2} = -\frac{1}{3\pi} = h_d(-3)$$

$$h_d(2) = \frac{1}{2\pi} \sin \pi = 0 = h_d(-2),$$

$$h_d(4) = \frac{1}{4\pi} \sin 2\pi = 0 = h_d(-4), \quad h_d(5) = \frac{1}{5\pi} \sin \frac{5\pi}{2} = \frac{1}{5\pi} = h_d(-5)$$

Assuming the window function,

$$w(n) = \begin{cases} 1, & \text{for } -5 \leq n \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

$$h(n) = h_d(n) \cdot w(n) = h_d(n)$$

We have

Therefore, the designed filter coefficients are given as:

$$h(0) = \frac{1}{2}, h(1) = \frac{1}{\pi} = h(-1), h(2) = 0 = h(-2), h(3) = -\frac{1}{3\pi} = h(-3),$$

$$h(4) = 0 = h(-4), h(5) = \frac{1}{5\pi} = h(-5)$$

The above coefficients correspond to a non-causal filter which is not realizable.
The realizable digital filter transfer function $H(z)$ is given by

$$H(z) = z^{-(N-1)/2} \left[h(0) + \sum_{n=1}^{(N-1)/2} h(n) [z^{-n} + z^n] \right] = z^{-5} \left[h(0) + \sum_{n=1}^5 h(n) [z^{-n} + z^n] \right]$$

$$= z^{-5} \left[h(0) + h(1) [z + z^{-1}] + h(3) [z^3 + z^{-3}] + h(5) [z^5 + z^{-5}] \right]$$

$$= h(5) + h(3)z^{-2} + h(1)z^{-4} + h(0)z^{-5} + h(1)z^{-6} + h(3)z^{-8} + h(5)z^{-10}$$

$$= \frac{1}{5\pi} - \frac{1}{3\pi} z^{-2} + \frac{1}{\pi} z^{-4} + \frac{1}{2} z^{-5} + \frac{1}{\pi} z^{-6} - \frac{1}{3\pi} z^{-8} + \frac{1}{5\pi} z^{-10}$$

Therefore, the coefficients of the realizable digital filter are:

$$h(0) = \frac{1}{5\pi} = h(10), h(1) = 0 = h(9), h(2) = -\frac{1}{3\pi} = h(8),$$

$$h(3) = 0 = h(7), h(4) = \frac{1}{\pi} = h(6), h(5) = \frac{1}{2}$$

EXAMPLE 9.7 A low-pass filter is to be designed with the following desired frequency response:

$$H_d(e^{j\omega}) = \begin{cases} e^{-j2\omega}, & -\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4} \\ 0, & \frac{\pi}{4} \leq |\omega| \leq \pi \end{cases}$$

Determine the filter coefficients $h(n)$ if the window function is defined as:

$$w(n) = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Also determine the frequency response $H(e^{j\omega})$ of the designed filter.

Solution: For the given filter with

$$H_d(\omega) = \begin{cases} e^{-j2\omega}, & -\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4} \\ 0, & \frac{\pi}{4} \leq |\omega| \leq \pi \end{cases}$$

The filter coefficients are given by

$$\begin{aligned} h_d(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{-j2\omega} e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{j\omega(n-2)} d\omega \\ &= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-2)}}{j(n-2)} \right]_{-\pi/4}^{\pi/4} = \frac{1}{\pi(n-2)} \left[\frac{e^{j(n-2)\frac{\pi}{4}} - e^{-j(n-2)\frac{\pi}{4}}}{2j} \right] \\ &= \frac{1}{\pi(n-2)} \sin(n-2) \frac{\pi}{4}, \quad n \neq 2. \end{aligned}$$

For $n = 2$, the filter coefficient can be obtained by applying L'Hospital rule to the above expression. Thus,

$$h_d(2) = \lim_{n \rightarrow 2} \frac{1}{\pi} \frac{\sin(n-2) \frac{\pi}{4}}{(n-2)} = \frac{1}{\pi} \cdot \frac{\pi}{4} = \frac{1}{4}$$

Since it is a linear phase filter, the other filter coefficients are given by

$$h_d(0) = \frac{1}{\pi(0-2)} \sin(0-2) \frac{\pi}{4} = \frac{1}{2\pi}$$

$$h_d(1) = \frac{1}{\pi(1-2)} \sin(1-2) \frac{\pi}{4} = \frac{1}{\sqrt{2}\pi}$$

$$h_d(3) = \frac{1}{\pi(3-2)} \sin(3-2) \frac{\pi}{4} = \frac{1}{\sqrt{2}\pi}$$

$$h_d(4) = \frac{1}{\pi(4-2)} \sin(4-2) \frac{\pi}{4} = \frac{1}{2\pi}$$

The filter coefficients of the filter using rectangular window would be then

$$h(n) = h_d(n) \cdot w(n) = h_d(n)$$

Therefore,
$$h(0) = \frac{1}{2\pi} = h(4), \quad h(1) = \frac{1}{\sqrt{2}\pi} = h(3), \quad \text{and} \quad h(2) = \frac{1}{4}$$

are the coefficients of the designed digital filter.

The realizable digital filter is:

$$\begin{aligned} H(z) &= \sum_{n=0}^{N-1} h(n) z^{-n} = h(0) + h(1) z^{-1} + h(2) z^{-2} + h(3) z^{-3} + h(4) z^{-4} \\ &= \frac{1}{2\pi} + \frac{1}{\sqrt{2}\pi} z^{-1} + 0.25 z^{-2} + \frac{1}{\sqrt{2}\pi} z^{-3} + \frac{1}{2\pi} z^{-4} \\ &= z^{-2} \left[0.25 + \frac{1}{\sqrt{2}\pi} (z + z^{-1}) + \frac{1}{2\pi} (z^2 + z^{-2}) \right] \end{aligned}$$

The frequency response $H(\omega)$ of the digital filter is given by

$$\begin{aligned} H(\omega) &= \sum_{n=0}^4 h(n) e^{-j\omega n} \\ &= h(0) + h(1) e^{-j\omega} + h(2) e^{-j2\omega} + h(3) e^{-j3\omega} + h(4) e^{-j4\omega} \\ &= e^{-j2\omega} [h(0) e^{j2\omega} + h(1) e^{j\omega} + h(2) + h(3) e^{-j\omega} + h(4) e^{-j2\omega}] \\ &= e^{-j2\omega} [h(2) + h(1) (e^{j\omega} + e^{-j\omega}) + h(0) (e^{j2\omega} + e^{-j2\omega})] \\ &= e^{-j2\omega} \left[\frac{1}{4} + \frac{\sqrt{2}}{\pi} \cos \omega + \frac{1}{\pi} \cos 2\omega \right] \end{aligned}$$

EXAMPLE 9.8 A filter is to be designed with the following desired frequency response

$$H_d(e^{j\omega}) = \begin{cases} 0, & -\frac{\pi}{2} \leq \omega \leq \frac{\pi}{2} \\ e^{-j2\omega}, & \frac{\pi}{2} < |\omega| \leq \pi \end{cases}$$

Determine the filter coefficients $h(n)$, if the window function is defined as

$$w(n) = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Also determine the frequency response $H(e^{j\omega})$ of the designed filter.

Solution: For the given high-pass filter with

$$H_d(\omega) = \begin{cases} 0, & -\frac{\pi}{2} \leq \omega \leq \frac{\pi}{2} \\ e^{-j2\omega}, & \frac{\pi}{2} \leq |\omega| \leq \pi \end{cases}$$

the filter coefficients are given by

$$\begin{aligned} h_d(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{-\pi/2} e^{-j2\omega} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{\pi/2}^{\pi} e^{-j2\omega} e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{-\pi/2} e^{j\omega(n-2)} d\omega + \frac{1}{2\pi} \int_{\pi/2}^{\pi} e^{j\omega(n-2)} d\omega \\ &= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-2)}}{j(n-2)} \right]_{-\pi}^{-\pi/2} + \frac{1}{2\pi} \left[\frac{e^{j\omega(n-2)}}{j(n-2)} \right]_{\pi/2}^{\pi} \\ &= \frac{1}{j2\pi(n-2)} \left[e^{-j(n-2)\frac{\pi}{2}} - e^{-j(n-2)\pi} + e^{j(n-2)\pi} - e^{j(n-2)\frac{\pi}{2}} \right] \\ &= \frac{1}{\pi(n-2)} \left\{ \left[\frac{e^{j(n-2)\pi} - e^{-j(n-2)\pi}}{2j} \right] - \left[\frac{e^{j(n-2)\frac{\pi}{2}} - e^{-j(n-2)\frac{\pi}{2}}}{2j} \right] \right\} \\ &= \frac{1}{\pi(n-2)} \left[\sin(n-2)\pi - \sin(n-2)\frac{\pi}{2} \right], \quad n \neq 2 \end{aligned}$$

For $n = 2$, $h_d(n)$ is indeterminate. So using L'Hospital rule, we have

$$h_d(2) = \lim_{n \rightarrow 2} \frac{1}{\pi} \left[\frac{\sin(n-2)\pi}{(n-2)} - \frac{\sin(n-2)\pi/2}{n-2} \right] = \frac{1}{\pi} \left[\pi - \frac{\pi}{2} \right] = 1 - \frac{1}{2} = \frac{1}{2}$$

The other filter coefficients are:

$$h_d(0) = \frac{1}{\pi(0-2)} \left[\sin(0-2)\pi - \sin(0-2)\frac{\pi}{2} \right] = 0$$

$$h_d(1) = \frac{1}{\pi(1-2)} \left[\sin(1-2)\pi - \sin(1-2)\frac{\pi}{2} \right] = -\frac{1}{\pi}$$

$$h_d(3) = \frac{1}{\pi(3-2)} \left[\sin(3-2)\pi - \sin(3-2)\frac{\pi}{2} \right] = -\frac{1}{\pi}$$

$$h_d(4) = \frac{1}{\pi(4-2)} \left[\sin(4-2)\pi - \sin(4-2)\frac{\pi}{2} \right] = 0$$

Since $h(n) = h_d(n) \cdot w(n)$, applying the window function, the new filter coefficients are:

$$h(0) = 0, h(1) = -\frac{1}{\pi}, h(2) = \frac{1}{2}, h(3) = -\frac{1}{\pi}, \text{ and } h(4) = 0$$

The transfer function of the filter is:

$$\begin{aligned} H(z) &= \sum_{n=0}^{N-1} h(n) z^{-n} \\ &= h(0) + h(1) z^{-1} + h(2) z^{-2} + h(3) z^{-3} + h(4) z^{-4} \\ &= 0 - \frac{1}{\pi} z^{-1} + \frac{1}{2} z^{-2} - \frac{1}{\pi} z^{-3} + 0 \\ &= z^{-2} \left[\frac{1}{2} - \frac{1}{\pi} (z + z^{-1}) \right] \end{aligned}$$

The frequency response $H(\omega)$ is given by

$$\begin{aligned} H(\omega) &= \sum_{n=0}^4 h(n) e^{-j\omega n} \\ &= h(0) + h(1) e^{-j\omega} + h(2) e^{-j2\omega} + h(3) e^{-j3\omega} + h(4) e^{-j4\omega} \\ &= -\frac{1}{\pi} e^{-j\omega} + \frac{1}{2} e^{-j2\omega} - \frac{1}{\pi} e^{-j3\omega} \\ &= e^{-j2\omega} \left[\frac{1}{2} - \frac{1}{\pi} (e^{j\omega} + e^{-j\omega}) \right] \\ &= e^{-j2\omega} \left[\frac{1}{2} - \frac{2}{\pi} \cos \omega \right] \end{aligned}$$

TABLE 9.1 The normalized ideal (desired) frequency response and impulse response for FIR filter design using windows. FIR Filters 675

Type of filter	Ideal (desired) frequency response	Ideal (desired) impulse response
Low-pass filter	$H_d(\omega) = \begin{cases} e^{-j\alpha\omega}; & -\omega_c \leq \omega \leq \omega_c \\ 0 & ; \quad -\pi \leq \omega < -\omega_c \\ 0 & ; \quad \omega_c < \omega \leq \pi \end{cases}$	$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\alpha} e^{j\omega n} d\omega$ $= \frac{\sin \omega_c (n - \alpha)}{\pi(n - \alpha)}$
High-pass filter	$H_d(\omega) = \begin{cases} e^{-j\alpha\omega}; & -\pi \leq \omega \leq -\omega_c \\ e^{-j\alpha\omega}; & \omega_c \leq \omega \leq \pi \\ 0 & ; \quad -\omega_c < \omega < \omega_c \end{cases}$	$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$ $= \frac{1}{2\pi} \int_{-\pi}^{-\omega_c} e^{-j\omega\alpha} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{\omega_c}^{\pi} e^{-j\omega\alpha} e^{j\omega n} d\omega$ $= \frac{\sin(n - \alpha)\pi - \sin \omega_c (n - \alpha)}{\pi(n - \alpha)}$
Band-pass filter	$H_d(\omega) = \begin{cases} e^{-j\omega\alpha}; & -\omega_{c2} \leq \omega \leq -\omega_{c1} \\ e^{-j\omega\alpha}; & \omega_{c1} \leq \omega < \omega_{c2} \\ 0 & ; \quad -\pi \leq \omega < -\omega_{c2} \\ 0 & ; \quad -\omega_{c1} < \omega < \omega_{c1} \\ 0 & ; \quad \omega_{c2} < \omega \leq \pi \end{cases}$	$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$ $= \frac{1}{2\pi} \int_{-\omega_{c2}}^{-\omega_{c1}} e^{-j\omega\alpha} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{\omega_{c1}}^{\omega_{c2}} e^{-j\omega\alpha} e^{j\omega n} d\omega$ $= \frac{\sin \omega_{c2} (n - \alpha) - \sin \omega_{c1} (n - \alpha)}{\pi(n - \alpha)}$
Band-stop filter	$H_d(\omega) = \begin{cases} e^{-j\omega\alpha}; & -\pi \leq \omega \leq -\omega_{c2} \\ e^{-j\omega\alpha}; & -\omega_{c1} \leq \omega \leq \omega_{c1} \\ e^{-j\omega\alpha}; & \omega_{c2} < \omega < \pi \\ 0 & ; \quad -\omega_{c2} \leq \omega \leq -\omega_{c1} \\ 0 & ; \quad \omega_{c1} < \omega < \omega_{c2} \end{cases}$	$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$ $= \frac{1}{2\pi} \int_{-\pi}^{-\omega_{c2}} e^{-j\omega\alpha} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{-\omega_{c1}}^{\omega_{c1}} e^{-j\omega\alpha} e^{j\omega n} d\omega$ $+ \frac{1}{2\pi} \int_{\omega_{c2}}^{\pi} e^{-j\omega\alpha} e^{j\omega n} d\omega$ $= \frac{\sin \omega_{c1} (n - \alpha) + \sin \pi (n - \alpha) - \sin \omega_{c2} (n - \alpha)}{\pi(n - \alpha)}$

EXAMPLE 9.10 Design a digital FIR low-pass filter using rectangular window by taking 9 samples of $w(n)$ and with a cutoff frequency of 1.2 rad/sec.

Solution: Cutoff frequency of given low-pass filter $\omega_c = 1.2$ rad/sec and $N = 9$. For a low-pass filter, the desired frequency response is

$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha}, & -\omega_c \leq \omega \leq \omega_c \\ 0, & \text{otherwise} \end{cases}$$

The desired impulse response is obtained by taking the inverse Fourier transform of $H_d(\omega)$. Therefore,

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\alpha} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)} d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)} \right]_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{\pi(n-\alpha)} \left[\frac{e^{j(n-\alpha)\omega_c} - e^{-j(n-\alpha)\omega_c}}{2j} \right]$$

$$= \frac{1}{\pi(n-\alpha)} \sin(n-\alpha)\omega_c \quad \text{for } n \neq \alpha$$

and for $n = \alpha$, $\frac{\sin(n-\alpha)\omega_c}{\pi(n-\alpha)}$ is $0/0$, which is indeterminate. Using L'hospital rule, we have

$$\text{for } n = \alpha, H_d(\alpha) = \lim_{n \rightarrow \alpha} \frac{\sin(n-\alpha)\omega_c}{\pi(n-\alpha)} = \frac{\omega_c}{\pi}.$$

The impulse response of FIR filter $h(n)$ is obtained by multiplying $h_d(n)$ by the window sequence. Therefore, Impulse response $h(n) = h_d(n) \cdot w_R(n)$.

$$\text{Rectangular window sequence, } w_R(n) = \begin{cases} 1, & \text{for } 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

$$\therefore h(n) = h_d(n); \quad \text{for } 0 \leq n \leq N-1$$

$$\text{Here } N = 9; \omega_c = 1.2 \text{ rad/sec and } \alpha = \frac{N-1}{2} = \frac{9-1}{2} = 4$$

$$\text{Therefore, we have } h(n) = \frac{\sin(n-4) \times 1.2}{\pi(n-4)}, \quad \text{for } n \neq 4$$

Therefore,

$$h(0) = \frac{\sin(0-4)1.2}{\pi(0-4)} = -0.0793, \quad h(1) = \frac{\sin(1-4)1.2}{\pi(1-4)} = -0.0470$$

$$h(2) = \frac{\sin(2-4)1.2}{\pi(2-4)} = 0.1075, \quad h(3) = \frac{\sin(3-4)1.2}{\pi(3-4)} = 0.2967$$

$$h(4) = \frac{\omega_c}{\pi} = \frac{1.2}{\pi} = 0.382, \quad h(5) = \frac{\sin(5-4)1.2}{\pi(5-4)} = 0.2967$$

$$h(6) = \frac{\sin(6-4)1.2}{\pi(6-4)} = 0.1075, \quad h(7) = \frac{\sin(7-4)1.2}{\pi(7-4)} = -0.047$$

$$h(8) = \frac{\sin(8-4)1.2}{\pi(8-4)} = -0.0793$$

Here we can observe that $h(0) = h(8)$, $h(1) = h(7)$, $h(2) = h(6)$ and $h(3) = h(5)$, i.e., the impulse response is satisfying the symmetry condition $h(N-1-n) = h(n)$.

The transfer function of the filter is given by

$$\begin{aligned}
 H(z) &= \sum_{n=0}^{N-1} h(n) z^{-n} \\
 &= h(0) + h(1)z^{-1} + h(2)z^{-2} + h(3)z^{-3} + h(4)z^{-4} + h(5)z^{-5} + h(6)z^{-6} + h(7)z^{-7} + h(8)z^{-8} \\
 &= z^{-4} [h(4) + h(3)[z + z^{-1}] + h(2)[z^2 + z^{-2}] + h(1)[z^3 + z^{-3}] + h(0)[z^4 + z^{-4}]] \\
 &= z^{-4} [0.382 + 0.2967[z + z^{-1}] + 0.1075[z^2 + z^{-2}] - 0.4070[z^3 + z^{-3}]] - 0.0793[z^4 + z^{-4}] \\
 &= -0.0793 - 0.4070z^{-1} + 0.1075z^{-2} + 0.2967z^{-3} + 0.382z^{-4} + 0.2967z^{-5} + 0.1075z^{-6} \\
 &\quad - 0.4070z^{-7} - 0.0793z^{-8}
 \end{aligned}$$

$$\begin{aligned}
 H(\omega) &= e^{-j\omega(N-1)/2} \left[h\left(\frac{N-1}{2}\right) + \sum_{n=1}^{(N-1)/2} 2h\left(\frac{N-1}{2} - n\right) \cos \omega n \right] \\
 &= e^{-j4\omega} \left[h(4) + \sum_{n=1}^4 2h(4-n) \cos \omega n \right] \\
 &= e^{-j4\omega} \{h(4) + 2h(3) \cos \omega + 2h(2) \cos 2\omega + 2h(1) \cos 3\omega + 2h(0) \cos 4\omega\} \\
 &= e^{-j4\omega} \{0.382 + 2 \times 0.2967 \cos \omega + 2 \times 0.1075 \cos 2\omega + 2 \times (-0.047) \cos 3\omega \\
 &\quad + 2 \times (-0.0793) \cos 4\omega\} \\
 &= e^{-j4\omega} \{0.382 + 0.5934 \cos \omega + 0.215 \cos 2\omega - 0.094 \cos 3\omega - 0.1586 \cos 4\omega\}
 \end{aligned}$$

The magnitude response is:

$$|H(\omega)| = 0.382 + 0.5934 \cos \omega + 0.215 \cos 2\omega - 0.094 \cos 3\omega - 0.1586 \cos 4\omega$$

EXAMPLE 9.11 Design a high-pass filter using Hamming window, with a cutoff frequency of 1.2 rad/sec and $N = 9$.

Solution: The desired frequency response $H_d(\omega)$ for a high-pass filter is

$$H_d(\omega) = \begin{cases} 0 & , -\omega_c \leq \omega \leq \omega_c \\ e^{-j\omega\alpha} & , \omega_c < |\omega| \leq \pi \end{cases}$$

The desired impulse response $h_d(n)$ is obtained by taking the inverse Fourier transform of $H_d(\omega)$.

$$\begin{aligned}
h_d(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j\omega\alpha} \cdot e^{j\omega n} d\omega \\
&= \frac{1}{2\pi} \int_{-\pi}^{-\omega_c} e^{j\omega(n-\alpha)} d\omega + \frac{1}{2\pi} \int_{\omega_c}^{\pi} e^{j\omega(n-\alpha)} d\omega \\
&= \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)} \right]_{-\pi}^{-\omega_c} + \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)} \right]_{\omega_c}^{\pi} \\
&= \frac{1}{2\pi} \left[\frac{e^{-j(n-\alpha)\omega_c} - e^{-j(n-\alpha)\pi} + e^{j(n-\alpha)\pi} - e^{j(n-\alpha)\omega_c}}{j(n-\alpha)} \right] \\
&= \frac{1}{\pi(n-\alpha)} \left[\frac{e^{j(n-\alpha)\pi} - e^{-j(n-\alpha)\pi}}{2j} - \frac{e^{j(n-\alpha)\omega_c} - e^{-j(n-\alpha)\omega_c}}{2j} \right] \\
&= \frac{1}{\pi(n-\alpha)} \{ \sin(n-\alpha)\pi - \sin(n-\alpha)\omega_c \}
\end{aligned}$$

When $n = \alpha$, the terms $\frac{\sin(n-\alpha)\pi}{\pi(n-\alpha)}$ and $\frac{\sin(n-\alpha)\omega_c}{\pi(n-\alpha)}$ become 0/0 which is indeterminate

Using L'Hospital rule, we have for $n = \alpha$.

$$\begin{aligned}
h_d(n) &= \lim_{n \rightarrow \alpha} \frac{1}{\pi} \left[\frac{\sin(n-\alpha)\pi}{(n-\alpha)} - \frac{\sin(n-\alpha)\omega_c}{(n-\alpha)} \right] \\
&= \frac{1}{\pi} [\pi - \omega_c] = 1 - \frac{\omega_c}{\pi}
\end{aligned}$$

For $n \neq \alpha$,

$$h_d(n) = \frac{\sin(n-\alpha)\pi - \sin(n-\alpha)\omega_c}{\pi(n-\alpha)}$$

Here $\alpha = \frac{N-1}{2} = \frac{9-1}{2} = 4$ is an integer and since n is also an integer, $(n-\alpha)$ is also integer and so $\sin(n-\alpha)\pi = 0$.

$$\therefore h_d(n) = -\frac{\sin(n-\alpha)\omega_c}{\pi(n-\alpha)} = -\frac{\sin(n-4)1.2}{\pi(n-4)}$$

Therefore,

$$\begin{aligned}
 h_d(0) &= -\frac{\sin(0-4)1.2}{\pi(0-4)} = 0.0792, & h_d(1) &= -\frac{\sin(1-4)1.2}{\pi(1-4)} = 0.0469 \\
 h_d(2) &= -\frac{\sin(2-4)1.2}{\pi(2-4)} = -0.1075, & h_d(3) &= -\frac{\sin(3-4)1.2}{\pi(3-4)} = -0.2966 \\
 h_d(4) &= 1 - \frac{\omega_c}{\pi} = 1 - \frac{1.2}{\pi} = 0.618, & h_d(5) &= -\frac{\sin(5-4)1.2}{\pi(5-4)} = -0.2966 \\
 h_d(6) &= -\frac{\sin(6-4)1.2}{\pi(6-4)} = -0.1075, & h_d(7) &= -\frac{\sin(7-4)1.2}{\pi(7-4)} = -0.0469 \\
 h_d(8) &= -\frac{\sin(8-4)1.2}{\pi(8-4)} = -0.0792
 \end{aligned}$$

The window sequence for Hamming window is given by

$$w_H(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right); \quad \text{for } n = 0 \text{ to } N-1$$

Therefore,

$$w_H(0) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 0}{9-1}\right) = 0.08$$

$$w_H(1) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 1}{9-1}\right) = 0.2147$$

$$w_H(2) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 2}{9-1}\right) = 0.54$$

$$w_H(3) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 3}{9-1}\right) = 0.8652$$

$$w_H(4) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 4}{9-1}\right) = 1$$

$$w_H(5) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 5}{9-1}\right) = 0.8652$$

$$w_H(6) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 6}{9-1}\right) = 0.54$$

$$w_H(7) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 7}{9-1}\right) = 0.2147$$

$$w_H(8) = 0.54 - 0.46 \cos\left(\frac{2\pi \times 8}{9-1}\right) = 0.08$$

The filter coefficients are $h(n) = h_d(n)w_H(n)$

$$\begin{aligned} \therefore h(0) &= h_d(0) w_H(0) = 0.0792 \times 0.08 = 0.0063 \\ h(1) &= h_d(1) w_H(1) = 0.0469 \times 0.2147 = 0.0100 \\ h(2) &= h_d(2) w_H(2) = -0.1075 \times 0.54 = -0.0580 \\ h(3) &= h_d(3) w_H(3) = -0.2966 \times 0.8652 = -0.2566 \\ h(4) &= h_d(4) w_H(4) = 0.618 \times 1 = 0.618 \\ h(5) &= h_d(5) w_H(5) = -0.2966 \times 0.8652 = -0.2566 \\ h(6) &= h_d(6) w_H(6) = -0.1075 \times 0.54 = -0.0580 \\ h(7) &= h_d(7) w_H(7) = 0.0469 \times 0.2147 = 0.0100 \\ h(8) &= h_d(8) w_H(8) = 0.0792 \times 0.08 = 0.0063 \end{aligned}$$

From the above calculations, we can observe that $h(N-1-n) = h(n)$, i.e., the impulse response is symmetrical with centre of symmetry at $n = 4$. The frequency response of the filter is:

$$\begin{aligned} H(\omega) &= \sum_{n=0}^{N-1} h(n) e^{-j\omega n} \\ &= \left[h(0) + h(1)e^{-j\omega} + h(2)e^{-j2\omega} + h(3)e^{-j3\omega} + h(4)e^{-j4\omega} + h(5)e^{-j5\omega} \right. \\ &\quad \left. + h(6)e^{-j6\omega} + h(7)e^{-j7\omega} + h(8)e^{-j8\omega} \right] \\ &= e^{-j4\omega} [h(4) + 2h(3) \cos \omega + 2h(2) \cos 2\omega + 2h(1) \cos 3\omega + 2h(0) \cos 4\omega] \\ &= e^{-j4\omega} \left[0.618 + 2(-0.256) \cos \omega + 2(-0.058) \cos 2\omega + 2(0.010) \cos 3\omega \right. \\ &\quad \left. + 2(0.0063) \cos 4\omega \right] \\ &= e^{-j4\omega} [0.618 - 0.5132 \cos \omega - 0.116 \cos 2\omega + 0.020 \cos 3\omega + 0.012 \cos 4\omega] \end{aligned}$$

The magnitude response is given by

$$|H(\omega)| = 0.618 - 0.5132 \cos \omega - 0.116 \cos 2\omega + 0.020 \cos 3\omega + 0.012 \cos 4\omega$$

The transfer function of the filter is given by

$$\begin{aligned}
 H(z) &= \sum_{n=0}^{N-1} h(n) z^{-n} = \sum_{n=0}^8 h(n) z^{-n} \\
 &= h(0) + h(1)z^{-1} + h(2)z^{-2} + h(3)z^{-3} + h(4)z^{-4} + h(5)z^{-5} + h(6)z^{-6} + h(7)z^{-7} + h(8)z^{-8} \\
 &= 0.0063 + 0.010z^{-1} - 0.0580z^{-2} - 0.2566z^{-3} + 0.618z^{-4} - 0.2566z^{-5} \\
 &\quad - 0.0580z^{-6} + 0.0100z^{-7} + 0.0063z^{-8} \\
 &= z^{-4} [0.618 - 0.2566(z + z^{-1}) - 0.0580(z^2 + z^{-2}) + 0.0100(z^3 + z^{-3}) + 0.0063(z^4 + z^{-4})]
 \end{aligned}$$

EXAMPLE 9.12 Design a band-pass filter to pass frequencies